

GRAHAMSTOWN, South Africa

WMO: 688490

Lat: 33.291S

Lon: 26.503E

Elev: 642

StdP: 93.85

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	5.4	-7.2	2.2	18.7	-4.9	2.7	17.1	13.1	19.8	11.7	17.4	3.6	290	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.3	32.4	19.1	30.3	18.8	28.3	18.5	21.9	27.5	21.1	26.2	20.4	25.1	4.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	16.3	23.4	19.6	15.5	22.5	19.0	14.9	21.9	67.1	27.5	64.2	26.2	61.6	25.1	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.0	8.0	DB	2.2	37.3	0.9	1.7	1.6	38.6	1.1	39.5	0.6	40.5	0.0	41.7	(4)
(5)			WB	1.1	23.3	1.0	0.7	0.4	23.9	-0.2	24.3	-0.8	24.7	-1.5	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	20.2	20.7	19.4	17.6	15.7	13.5	13.3	13.9	14.4	15.9	17.1	18.7	(6)	
	DBStd	4.25	3.07	3.20	3.38	3.72	3.28	3.28	3.65	3.65	3.51	3.69	3.27	3.18	(7)	
	HDD10.0	30	0	0	0	1	1	8	11	6	3	1	0	0	(8)	
	HDD18.3	975	16	11	27	58	94	148	158	146	127	95	62	33	(9)	
	CDD10.0	2469	317	300	291	229	179	112	114	126	136	183	212	270	(10)	
	CDD18.3	371	74	77	59	36	13	2	3	8	9	19	24	44	(11)	
	CDH23.3	2429	436	472	397	206	76	10	21	63	91	181	177	299	(12)	
(13)	CDH26.7	776	142	170	141	58	9	0	1	12	24	65	50	104	(13)	
(14) Wind	WSAvg	3.9	4.0	3.8	3.4	3.4	3.5	4.0	4.1	3.9	3.9	4.1	4.1	4.1	(14)	
(15) Precipitation	PrecAvg	632	54	57	61	55	29	37	35	49	51	70	72	62	(15)	
	PrecMax	863	99	116	129	140	109	146	123	213	174	188	267	238	(16)	
	PrecMin	386	22	16	7	9	6	2	1	3	12	15	6	9	(17)	
	PrecStd	132	19	27	27	35	26	36	34	51	42	39	53	55	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	34.5	34.1	31.6	27.9	24.6	25.5	28.4	30.1	32.3	32.1	33.6	(19)	
		MCWB	20.2	20.1	19.1	17.9	14.6	12.0	11.7	12.2	14.1	16.2	19.4	19.0	(20)	
	2%	DB	30.3	31.1	30.7	28.6	25.4	22.6	23.3	25.0	25.9	28.3	28.2	29.8	(21)	
		MCWB	20.6	20.6	18.6	17.2	13.8	11.1	11.1	12.1	14.1	16.6	18.1	19.3	(22)	
	5%	DB	27.6	28.4	27.6	25.7	23.2	20.8	21.4	22.4	22.9	24.8	25.2	26.5	(23)	
		MCWB	20.0	20.1	18.7	16.9	13.5	10.7	10.5	11.6	13.6	16.0	17.6	18.8	(24)	
	10%	DB	25.4	26.1	25.1	23.2	20.9	18.7	19.2	19.7	20.4	22.1	22.7	24.1	(25)	
		MCWB	19.3	20.0	18.4	16.6	13.0	10.4	10.2	11.1	13.1	15.2	17.1	18.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	22.6	22.3	20.2	17.3	14.5	14.1	15.7	17.5	19.7	22.1	21.7	(27)	
		MCDB	29.2	28.5	27.8	25.3	21.5	17.9	19.1	19.9	22.8	26.5	24.9	28.7	(28)	
	2%	WB	21.5	21.8	20.9	19.1	15.7	13.2	13.0	14.5	16.1	18.1	19.9	20.7	(29)	
		MCDB	27.2	27.3	25.5	23.6	20.5	18.0	18.0	18.7	21.3	23.6	25.7	26.2	(30)	
	5%	WB	20.6	21.1	20.0	18.2	14.9	12.3	12.1	13.7	15.1	17.1	18.6	19.8	(31)	
		MCDB	25.5	25.9	24.6	22.7	19.7	16.9	16.8	17.5	19.3	22.1	23.2	24.3	(32)	
	10%	WB	19.8	20.4	19.1	17.4	14.2	11.5	11.4	12.7	14.2	16.1	17.6	18.8	(33)	
		MCDB	24.1	24.6	23.3	21.6	18.8	16.2	16.1	16.8	17.9	20.5	21.2	22.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.3	9.3	9.1	9.5	9.7	9.8	10.0	9.9	9.7	9.4	9.1	(35)	
		MCDBR	13.8	14.2	14.6	13.3	12.9	12.4	12.7	14.3	15.2	15.5	14.5	14.3	(36)	
	5% WB	MCWBR	5.0	4.7	4.5	4.3	4.3	4.5	4.4	4.6	5.1	5.5	5.8	5.5	(37)	
		MCWBR	11.6	11.6	11.1	10.6	10.3	9.8	9.8	10.0	11.4	12.0	12.1	11.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.360	0.362	0.347	0.324	0.308	0.294	0.297	0.337	0.359	0.369	0.352	0.356	(40)	
		taud	2.491	2.494	2.512	2.551	2.565	2.577	2.563	2.443	2.378	2.385	2.470	2.494	(41)	
	Ebn at Noon		978	955	933	903	866	855	869	872	905	936	979	985	(42)	
		Edh at Noon	115	111	102	87	76	70	75	95	114	122	117	116	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.82	6.22	5.22	4.06	3.17	2.83	3.03	3.80	4.92	5.80	6.63	7.10	(44)	
	RadStd		0.33	0.30	0.39	0.21	0.12	0.15	0.17	0.24	0.33	0.44	0.38	0.46	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (8 neighbors)	+0.35	+0.42	N/A	N/A	N/A	N/A	N/A	-85	+142	+45	(47)

Nomenclature: See separate page